Genes Linked to Type 2 Diabetes (T2D)

From the Top 15 Predictive Genes in a Logistical Regression Classifier

PHF2

Srinivasan, S., Chen, L., Todd, J., Divers, J., Gidding, S., Chernausek, S., … & Florez, J. C. (2021). *The first genome-wide association study for type 2 diabetes in youth: The Progress in Diabetes Genetics in Youth (ProDiGY) Consortium*. **Diabetes, 70**(4), 996–1005. DOI: 10.2337/db20-0443 (Identified a PHF2 variant associated with T2D)

Okuno, Y., Ohtake, F., Igarashi, K., Kanno, J., Matsumoto, T., … & Kato, S. (2013). *Epigenetic regulation of adipogenesis by PHF2 histone demethylase*. **Diabetes, 62**(5), 1426–1434. DOI: 10.2337/db12-0628 (Demonstrated that PHF2 regulates fat-cell differentiation, linking it to metabolic control)

GCK

Froguel, P., Vaxillaire, M., Sun, F., Velho, G., Zouali, H., Butel, M. O., … & Cohen, D. (1992). *Close linkage of glucokinase locus on chromosome 7p to early-onset non-insulin-dependent diabetes mellitus*. **Nature, 356**(6365), 162–164. DOI: 10.1038/356162a0 (Provided the first evidence that GCK mutations are linked to an inherited form of T2D)

Gidh-Jain, M., Takeda, J., Xu, L., Langefeld, C. D., Rich, S. S., Freedman, B. I., … & Permutt, M. A. (1993). *Nonsense mutation in the glucokinase gene causes early-onset type II diabetes mellitus*. **Proc. Natl. Acad. Sci. USA, 90**(13), 5864–5868. DOI: 10.1073/pnas.90.13.5864 (Identified a specific inactivating mutation in GCK that causes early-onset T2D)

SLC30A8

Sladek, R., Rocheleau, G., Rung, J., Dina, C., Shen, L., Serre, D., … & Froguel, P. (2007). *A genome-wide association study identifies novel risk loci for type 2 diabetes*. **Nature, 445**(7130), 881–885. DOI: 10.1038/nature05616 (Identified SLC30A8 as a T2D-associated gene in a GWAS)

Zeggini, E., Weedon, M. N., Lindgren, C. M., Voight, B. F., Saxena, R., … & Frayling, T. M. (2008). *Meta-analysis of genome-wide association data and large-scale replication identifies additional susceptibility loci for type 2 diabetes*. **Nature Genetics, 40**(5), 638–645. DOI: 10.1038/ng.120 (Replicated the SLC30A8 association and identified other T2D loci)

IRS1

Rung, J., Cauchi, S., Albrechtsen, A., Shen, L., Rocheleau, G., Balkau, B., … & Sladek, R. (2009). *Genetic variant near IRS1 is associated with type 2 diabetes, insulin resistance and hyperinsulinemia*. **Nature Genetics, 41**(10), 1110–1115. DOI: 10.1038/ng.443 (Identified and characterized an IRS1-linked variant influencing T2D risk and insulin resistance)

Yiannakouris, N., Cooper, J. A., Shah, S., Drenos, F., Ireland, H. A., Stephens, J. W., … & Humphries, S. E. (2012). *IRS1 gene variants, dysglycemic metabolic changes and type 2 diabetes risk*. **Diabetes, 61**(8), 2180–2184. DOI: 10.2337/db11-1456 (Examined metabolic effects of IRS1 variants, providing further evidence of their role in T2D via insulin resistance)

SREBF1

Grarup, N., Stender-Petersen, K. L., Andersson, E. A., Jørgensen, T., Borch-Johnsen, K., Hansen, T., & Pedersen, O. (2008). *Association of variants in the sterol regulatory element-binding factor 1 (SREBF1) gene with type 2 diabetes, glycemia, and insulin resistance in 15,734 Danish subjects*. **Diabetes, 57**(4), 1136–1142. DOI: 10.2337/db07-1534 (Demonstrated that multiple SREBF1 polymorphisms are associated with modestly increased T2D risk and hyperglycemia)

Kantartzis, K., Machann, J., Schick, F., Fritsche, A., Häring, H. U., & Stefan, N. (2009). *The SREBF-1 locus is associated with plasma adiponectin and hepatic insulin resistance*. **Diabetologia, 52**(4), 567–572. DOI: 10.1007/s00125-008-1245-x (Provided supporting evidence that variation in SREBF1 influences insulin resistance, a T2D risk factor)

KCNJ11

Gloyn, A. L., Weedon, M. N., Owen, K. R., Turner, M. J., Knight, B. A., Hattersley, A. T., … & Frayling, T. M. (2003). *Large-scale association studies of variants in genes encoding the β-cell KATP channel subunits Kir6.2 (KCNJ11) and SUR1 (ABCC8) confirm that the KCNJ11 E23K variant is associated with type 2 diabetes*. **Diabetes, 52**(2), 568–572. DOI: 10.2337/diabetes.52.2.568 (Provided strong evidence that the common E23K variant in KCNJ11 increases T2D risk)

Nielsen, E. M., Hansen, L., Carstensen, B., Echwald, S. M., Drivsholm, T., Glümer, C., … & Pedersen, O. (2003). *The E23K variant of Kir6.2 associates with impaired post-OGTT insulin secretion and increased risk of type 2 diabetes*. **Diabetes, 52**(2), 573–577. DOI: 10.2337/diabetes.52.2.573 (Showed that E23K carriers have impaired insulin response, explaining the variant's diabetogenic effect)

CDKAL1

Steinthorsdottir, V., Thorleifsson, G., Reynisdottir, I., Benediktsson, R., Jonsdottir, T., … & Stefansson, K. (2007). *A variant in CDKAL1 influences insulin response and risk of type 2 diabetes*. **Nature Genetics, 39**(6), 770–775. DOI: 10.1038/ng2043 (Original report identifying the CDKAL1 variant associated with T2D and noting its effect on insulin secretion)

Zeggini, E., Scott, L. J., Saxena, R., Voight, B. F., Marchini, J. L., Hu, T., … & Sladek, R. (2008). *Meta-analysis of genome-wide association data and large-scale replication identifies additional susceptibility loci for type 2 diabetes*. **Nature Genetics, 40**(5), 638–645. DOI: 10.1038/ng.120 (Confirmed the association of CDKAL1 with T2D in a large multi-cohort analysis)

JAZF1

Zeggini, E., Scott, L. J., Saxena, R., Voight, B. F., Marchini, J. L., Hu, T., … & McCarthy, M. I. (2008). *Meta-analysis of genome-wide association data and large-scale replication identifies additional susceptibility loci for type 2 diabetes*. **Nature Genetics, 40**(5), 638–645. DOI: 10.1038/ng.120 (One of the first studies to identify JAZF1 as a T2D risk locus)

Bouatia-Naji, N., Rocheleau, G., Van Lommel, L., Lemaire, K., Schuit, F., … & Froguel, P. (2010). *A polymorphism within the **JAZF1** gene associates with an enhanced pancreatic β-cell function and a reduced type 2 diabetes risk in humans*. **Cell Metabolism, 11**(5), 360–368. DOI: 10.1016/j.cmet.2010.03.010 (Explored functional impact of JAZF1 variation on beta-cell function, providing biological context for the GWAS finding)

GLIS3

Cho, Y. S., Chen, C. H., Hu, C., Long, J., Ong, R. T., Sim, X., … & Tai, E. S. (2012). *Meta-analysis of genome-wide association studies identifies eight new loci for type 2 diabetes in East Asians*. **Nature Genetics, 44**(1), 67–72. DOI: 10.1038/ng.1019 (Reported GLIS3 as a novel T2D-associated locus in East Asian cohorts)

Meulebrouck, S., Scherrer, V., Boutry, R., Toussaint, B., Vaillant, E., Vaxillaire, M., … & Bonnefond, A. (2024). *Pathogenic monoallelic variants in GLIS3 increase type 2 diabetes risk and identify a subgroup of patients sensitive to sulfonylureas*. **Diabetologia, 67**(2), 327–332. DOI: 10.1007/s00125-023-06035-x (Demonstrated that individuals with rare deleterious GLIS3 variants have higher T2D risk, linking GLIS3 dysfunction to adult-onset T2D)

ZNRF3

Spracklen, C. N., Horikoshi, M., Kim, Y. J., Lin, K., Bragg, F., Moon, S., … & Mohlke, K. L. (2020). *Identification of type 2 diabetes loci in 433,540 East Asian individuals*. **Nature, 582**(7811), 240–245. DOI: 10.1038/s41586-020-2263-3 (Identified ZNRF3 as a genome-wide significant locus for T2D in East Asian populations)

Vujkovic, M., Keaton, J. M., Lynch, J. A., Miller, D. R., Zhou, J., … & Wilson, J. G. (2020). *Discovery of 318 new risk loci for type 2 diabetes and related vascular outcomes among 1.4 million participants in a multi-ancestry meta-analysis*. **Nature Genetics, 52**(7), 680–691. DOI: 10.1038/s41588-020-0637-y (Massive meta-analysis across multiple ethnicities; reported ZNRF3 as one of the new T2D risk genes, implicating Wnt signaling in diabetes)